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Stored Procedure :-

Stored Procedure is a group of SQL statements stored in a database that we can execute whenever we need them. Instead of writing the same SQL query again & again, we can save it as a procedure & call it whenever required.

Real-world example :

> Banking → Retrieve customer account details.
> E-Commerce → Calculate the total sales.
> HR → Retrieve employees information.
> Finance → Calculate monthly salaries.

Why do we use Stored Procedure ?

> Code Reusability - Write SQL code once & execute it multiple times.
> Time Saving - Avoid writing repeated queries.
> Better Organization - Keep multiple SQL statements together.
> Security - Access can be granted to execute a procedure without granting direct access to underlying tables.
> Performance - Reduce repeated client-server communication & allow database execution plans to be reused in many cases.
> Easy maintenance - Update the procedure in one place instead of changing multiple applications.

Syntax -

```
DELIMITER //
CREATE PROCEDURE procedure_name()
BEGIN
    -- SQL statements
END //
DELIMITER ;
```

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(*) DELIMITER // ⇒ changes the statement-ending symbol temporarily.

(*) DELIMITER ⇒ Restores the normal delimiter.

(*) Why do we use DELIMITER?
Normally, SQL statements end with a semicolon (;)
But a stored procedure can contain multiple SQL statements, each ending with a semicolon.
So, we temporarily change the delimiter to //, to tell MySQL where the complete procedure ends.

(*) Example -
```
DELIMITER //
CREATE PROCEDURE Employees()
BEGIN
    SELECT * FROM emp;
END //
DELIMITER ;
```

(*) How to execute the procedure - CALL Employees();

(*) Types of Parameters in Stored Procedures :-
Stored Procedures support three types of parameters -

1. IN Parameter ⇒ Used to pass a value into the procedure.
Example - Pass a department name & retrieve employees from that department. (Input)
Retrieve employees belonging to a particular department.

```
DELIMITER //
CREATE PROCEDURE EmployeesByDept(IN dept_name VARCHAR(50))
BEGIN
    SELECT * FROM emp
    WHERE department = dept_name;
END //
DELIMITER ;
```

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CALL EmployeesByDept('IT'); → Execute.

> We pass 'IT' as Input & the procedure retrieves employees working in the IT department.

2. OUT Parameter ⇒ Used to return a value from the procedure. (Output)

Example - Return the total number of Employees.

```
DELIMITER //
CREATE PROCEDURE countEmployees( OUT total INT)
BEGIN
    SELECT COUNT(*)
    INTO total
    FROM employees Emp;
END//
DELIMITER ;
```

Execute → CALL countEmployees(@emp_count);
SELECT @emp_count.

3. INOUT Parameter ⇒ (Input + Output) Used to accept an input value, modify it inside the procedure, & return the modified value.

Example - Add 1000 to an employee's salary amount passed into the procedure.

```
DELIMITER //
CREATE PROCEDURE IncreaseAmount( INOUT amount
        DECIMAL(10,2))
BEGIN
    SET amount = amount + 1000;
END//
DELIMETER ;
```

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Execute ⇒ SET @Salary = 30000;
CALL IncreaseAmount (@Salary);
SELECT @Salary;

> We pass 30000 into the procedure.
> The procedure adds 1000.
> The updated value is returned through the same variable.

(*) Local Variables in Stored Procedures :- A local variable is a temporary variable declared inside a stored procedure to store a value.
We use the 'DECLARE' Keyword.
Syntax : DECLARE variable_name datatype;

(*) Conditional Statements in Stored Procedures :- We can use conditional statements such as IF, ELSEIF, and ELSE inside stored procedures.

(*) LOOPS in Stored Procedures :- Loops are used to execute a set of statements repeatedly.
MySQL supports -
> LOOP → Repeats statements until we explicitly exit using 'LEAVE'.
> WHILE → Executes statements as long as the condition is true.
> REPEAT → Executes statements first & checks the condition afterward.

Loop — How it works.
> WHILE - Checks the condition before execution.
> REPEAT - Checks the condition after execution.
> LOOP - Continues until explicitly stopped.

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Task	SQL command.
> Create	CREATE PROCEDURE procedure_name()
> Execute	CALL procedure_name()
> View procedures	SHOW PROCEDURE STATUS.
> View procedure definition	SHOW CREATE PROCEDURE procedure_name.
> Delete	DROP PROCEDURE procedure_name.

(*) Important Questions :-

1. What is a stored procedure in SQL.
2. Why do we use stored procedure.
3. What is the difference between a stored procedure & a normal SQL query.
4. What is the purpose of the DELIMITER command.
5. What is the difference between IN, OUT and INOUT parameters.
6. How do you execute a stored procedure.
7. Can a stored procedure contain multiple SQL statements.
8. What is the difference between local variables & session variables.
9. Can we use IF ELSE and loops inside stored procedures.
10. How do you delete a stored procedure.